



Stoned or boiled lobsters

A Maine restaurant owner wants to drug the lobsters first since boiling them alive seems a inhumanly feat. <https://digit.in/oct18-71-1>



AI gliders

Eagles soar without flapping their wings for long periods of time. Scientists are using AI gliders to learn how <https://digit.in/oct18-71-2>

in the ocean, and captures any litter in its way. This is essentially dispatching an artificial coastline into the ocean, where there is none to stop the floating garbage. The floater is shaped like a U so that all the garbage gets concentrated in its center. The floater juts out of the water just enough to prevent plastic from going over it. The skirt below captures the bulk of the plastic, which floats just beneath the surface. The skirt is continuous, and not a net. This ensures that even small pieces of plastic, that float in the water, are trapped. At the same time, the skirt prevents marine life from getting entangled or tagged, they can simply swim under or around it. Additionally, it ensures that the current flows beneath the skirt, on which the tiny marine life, without the ability to swim, can hitch a ride.

There is no artificial propulsion on board, and the floater makes its way through the oceans riding on the natural currents, the wind, and the waves. The trajectory of the floater is calculated by algorithms ahead of the deployment. There is an energy system on board to capture solar energy for the lights and on board Automatic Identification System – a satellite based system to keep track of vessels in the ocean. The system prevents ocean vehicles from running into the floater. There are also two high definition 360 degrees cameras in the middle of the floater, for remote real time monitoring. There are lanterns and navigational beacons to alert any vehicles in the area. However, there is not much traffic in the great Pacific garbage patch anyway.

The floater and the skirt are built to be long lasting and robust, to endure any conditions that the sea can throw at it. The engineers considered extreme cases that occur only once in a hundred years. The main reason for the hardness of the floater is how flexible it is. Even when faced with five meter high waves, the floater simply floats over these waves. A system of sensors along the entire length of the floater makes sure that it is functioning as expected at all times.

The garbage is cleaned up periodically by oceanic “garbage trucks” – ships that visit the floater and collect the garbage trapped by the floater. The actual collection is a manual process, which ensures that there is no marine life that is picked up along with the garbage. The collected plastic waste is brought back to land, and sold to recyclers. Time Magazine called the “ocean vacuum cleaner” one of the best inventions of 2015.

THE IMPACT

The System 001 is the first operation by the Ocean Cleanup. The initial plan was to create a moored barrier, permanently anchored in the ocean. A feasibility study showed that a moving floater would allow the ocean to clean itself up. The next step was to raise funds through crowdfunding on the ABN Amro's Seeds platform. A total

shape as it makes its way through the ocean. Wilson also needs to move at a particular speed to make sure that it traps the plastic as intended. One of the objectives of the trial period is to monitor the speed, to ensure future deployments have the necessary speeds. Boats will also troll Wilson and orient it in all sorts of directions, to make sure that the floater reorients itself in the direction of the garbage. Another parameter of the test is to check that Wilson will not get damaged during the period of the operation. After the two weeks of the Pacific trial, Wilson will travel the 800 nautical miles towards the Pacific Garbage Patch.

The September 8 launch is just the beginning of the cleanup operation of the Pacific garbage patch. The floater is scalable, and eventually the organisation wants to deploy 100



How the floater behaves in rough seas

of US \$2,154,282 was raised, making it one of the most successful non-profit fundings. There were a number of tests in the North Atlantic and the North Sea, and the design was improved over these iterations. On September 8, the first cleanup operation of the System 001, nicknamed “Wilson” was successfully launch from the San Francisco Bay. Wilson travelled 350 nautical miles on its own, before embarking on a period of testing, known as the Pacific trials.

The two weeks of Pacific trials is to make sure Wilson maintains its U

kilometer long floaters. The skirts on these floaters will continue to have a length of 3 meters. In 2020, the team plans to move up to tackling garbage in the oceans worldwide. The eventual aim is to reduce up to 90 percent of the plastic waste in the ocean by the year 2040, and completely remove the plastic by 2050. Going by simulations, a rollout of the 100 kilometer long floaters would clean up 50 percent of the plastic in the Pacific Garbage patch as early as 2023. Now, wouldn't that be a wonderful day for a boat ride? 🚤